Applicant View

Complete First Academic Evaluator and Faculty Research Host Information

*Will the writer of your Academic Evaluation also be serving as your Faculty Research Host (this person must be a Harvard-affiliated faculty member)? This will determine the number of requests you will need to generate through the application system. While Academic Evaluations can be written by anyone, research positions can only be confirmed by your Faculty Research Host. If you would like for a post-doc or graduate student in a lab to write your Academic Evaluation, then please select the NOT option and solicit a separate Faculty Research Host Confirmation from your faculty research host/Principal Investigator in the "Request" section. Graduate students and postdoctoral fellows are not members of the Harvard faculty and cannot confirm your lab placement in this application.	My faculty research host will also be my Academic Evaluator.
*Do you already have a confirmed Harvard-affiliated faculty research host or are you still identifying a host? If you have not identified a faculty research host by February 1, please be sure to solicit a reference per the application instructions for the internal deadline and return to the application to request the faculty research host confirmation letter once you have found your lab host (must receive before March 1).	Yes, I have a confirmed faculty research host.

Supplementary Questions

SPUDS. Applicants whose applications involve data science can have their materials forwarded for consideration for the Summer Program in Undergraduate Data Science (SPUDS), another Research Village Program. Does your program involve data science and would you like for it to also be considered for SPUDS? If you indicate "yes" to both, you would not need to forward a separate application for SPUDS.	My project involves data science, please forward my application for consideration for SPUDS.

Research Sophistication

Our goal is to create a balanced PRISE cohort comprised of students with different levels of research experience. Please evaluate your own level of research experience using the drop-down options below and then share a bit more detail about your experiences with us in 1-2 sentences in the text entry box immediately following. Note well that your research experience need not have only taken place at Harvard. We have briefly outlined the different categories below (beginner, intermediate, advanced). There is no right or wrong answer; use your best judgment! We look forward to reading more about you!

- **Beginner:** little to no research experience; ex. you may have taken an introductory course with a lab component, but have not yet had any (or very little) non-academic hands-on research experience.
- **Intermediate:** 1-2 research experiences; ex: you have perhaps conducted 1-2 summers of full-time research during high school and/or college, some term-time research at Harvard, and/or have experience as a Research Assistant.
- **Advanced:** have had 2+ substantive research experiences during college; ex. you have conducted research over several summer(s) and term(s), perhaps even designing and making progress on a thesis or capstone project.

*What do you believe is your level of research sophistication?	Intermediate

*Please share 1-2 sentences explaining your research sophistication self-evaluation. (25 words)	Previous research includes involvement with the Stubbs group for Summer and Fall 2025, along with groups in computational sociology and physical chemistry during high school.
*Prior HSURV Experience. Have you previously participated in a Research Village program? If so, please indicate which one(s).	No prior Research Village program experience.

Applicant Information

*Harvard University ID	31712289
*First Name	Cyrus
*Last Name	Urheim
*Harvard University Email	cyrusurheim@college.Harvard.edu
*Anticipated Graduation Year	2028
*House	Mather House
*Primary Concentration	CHEMISTRY AND PHYSICS
*Concentration/Secondary Field/Joint Concentration	COMPUTER SCIENCE
*Current GPA	3.967
*Are you currently in good academic and disciplinary standing? You must be in good standing to apply.	Yes, I am in good standing.
*Citizenship Information (this information is collected for the purposes of determining payment type and in some instances, eligibility for some awards)	US Citizen
*State of Permanent Residence	Indiana

Applicant History

*Have you previously received any research awards from the Office of Undergraduate Research and Fellowships?	Yes
*Which awards have you received? Please check all that are relevant.	HCRP Summer
*What was the term in which you received the most recent of these awards?	Summer
*What was the most year in which you received the most recent award?	2025
*How much funding have you received from URAF in total?	$2,500.00

Applicant Demographic Information

This information is collected to help us understand patterns and trends in the community of students our office serves. Providing it is optional. It will not be shared with reviewers.

*What is your gender?	Man
What is your race/ethnicity?	Middle Eastern or North African White

Confirmed Project Information

*Project Title (5-15 words). Please capitalize your title using the conventions of your discipline. See https://capitalizemytitle.com/.	Sonar for Astrophysics: BATPING (Bidirectional Acoustic Time Pulser for Index Gradients)

*Please provide a brief summary of your project that details for a general audience the purpose, significance, methods, and anticipated outcomes (including how you plan to share your results) of your research or creative endeavor. (Maximum 100 words)

I will carry out research in experimental astrophysics under the direction of Prof. Christopher Stubbs, on instrumentation and data analysis to improve image quality for the Legacy Survey of Space and Time (LSST) at the newly commissioned Rubin Observatory. With its 63-cm diameter focal plane and 3.2-Gigapixel camera, LSST will advance our understanding of dark matter and dark energy in the universe. I will develop an acoustics-based system we call BATPING to measure temperature variations within the telescope dome responsible for image distortions. A key goal is the design of Machine-Learning based software to analyze BATPING's noisy acoustic signals.

*Primary Project Field: This will help us identify appropriate reviewers.	Astrophysics
*Project Faculty Advisor's First Name (do not include titles such as Prof., Dr., etc):	Christopher
* Project Faculty Advisor's Last Name (do not include post-nominals such as Ph.D.):	Stubbs
*Project Faculty Advisor's Official Harvard Email (you will also enter this in the formal request section)	stubbs@g.harvard.edu
*Project Faculty Advisor's Primary Harvard Affiliation	Faculty of Arts and Sciences (Harvard College, Graduate School of Arts and Sciences, SEAS, Continuing Education)
*Project Faculty Advisor's Department	Physics
*Are you currently a senior conducting thesis research AND is the project your senior thesis research?	No, I am not.
*Are you receiving course credit for this research? If so, you are ineligible for a stipend but must provide a clear budget of anticipated research-related expenses.	No, I am not conducting resesarch for credit

Ethical Considerations, Human Subjects Safety Information

Undergraduate research conducted by Harvard University students must meet the highest ethical and safety standards. Harvard College students considering a research project that includes interaction with and gathering or using any data sets about living people must submit their proposals for review to the Committee on the Use of Human Subjects (CUHS), which is the Institutional Review Board (IRB) for the Faculty of Arts and Sciences. Undergraduates conducting human subjects research are required to receive training by the IRB prior to undertaking the project. You may not begin your research - nor are we permitted to release any research funding to you - until your project has been cleared by the IRB!

Undergraduates conducting human subjects research must:

- Complete a training in the ethical conduct of research. This can be completed by attending a training session offered by Harvard or by obtaining CITI or NIH certification online. [You need not share proof of training.]
- Receive a decision on their research project from the IRB via the Committee on the Use of Human Subjects.

As you conduct research, please be sure to consult information provided by the Institutional Review Board through the Undergraduate Research Training Program about the ethical conduct of research, particularly as it pertains to Human Subjects research. It may take some time for the evaluation of your research protocols by colleagues in that office to take place. We strongly encourage applicants for our opportunities to move through the stages of their URTP Decision Form at the same time as they are navigating the submission of their research proposal; this will ensure that should your project be funded, you will be able to carry it out in the timeline articulated in your proposal.

Does your project involve human subjects? Check out the URTP Student Guide (PDF), which is an overview of the regulations that govern human subject research.	No, it does not involve human subjects.

Application Attachments

Application Attachments

Please upload your application materials here as instructed. The application system will compile your application into a single PDF file for review by our internal committee. Please be sure to ensure the legibility of all documents.

*Research Interests Essay. 500 words maximum, 12 point font, Times New Roman, double-spaced, in header please writer your name and indicate what essay this is.	Harvard PRISE 2026 Research Essay.pdf
*Community Interests Essay. 500 words maximum, 12 point font, Times New Roman, double-spaced, in header please writer your name and indicate what essay this is.	Harvard PRISE 2026 Community Essay.pdf
*Resume/CV (Maximum 1 page)	cyrusurheim_cv_02-01-26.pdf
*Current Transcript (this can be unofficial, but it must include the current term). If you have difficulty uploading this as a PDF due to encryption, you can export the file to a JPEG and upload that.	urheim_transcript_02-01-26.jpg

Scholarship and Fellowship Nomination Waiver and Agreement

Provisions Regarding Additional Funding

Students must note in their application if they have applied to other funding sources, and notify the Selection Committee immediately if they have had success in securing such funding.

Students who fail to notify the Committee of having applied to or received funding from other sources may be disqualified from receiving internal Office of Undergraduate Research and Fellowships Awards in the future.

Harvard College Scholarship Application and Nomination Permission Form and Waiver

1) I understand that it is an honor and privilege to be considered for Harvard College's nomination or review for internal awards and/or external scholarships. By my signature below, I acknowledge that certain information about me will be required and considered as part of the award competition process as detailed below.

2) I hereby waive my right to view faculty and institutional letters of recommendation or endorsement that are written for the purpose of this award competition. While copies of these letters may be provided to me by the authors, I understand this is done as a courtesy by the author and does not affect this waiver.

3) As required under the Family Educational Rights and Privacy Act, commonly known as FERPA, I hereby give permission for my grade point average (GPA), transcripts, and individual course grades to be used and discussed as a part of faculty and institutional letters of recommendation and endorsements and the applications for this award competition.

4) I grant permission for Harvard College to use my biographical information to publicize my nomination and/or receipt of this award, scholarship or fellowship. I also consent to have a copy of my application and all supporting materials retained indefinitely by the Office of Undergraduate Research and Fellowships or its successor and understand that it may be made available to future Harvard applicants as an example for them to review as they prepare their own applications.

5) I attest that all of the information I include in my application for this award, including listings of activities and awards, research undertaken or planned, and personal statements or other essays, are my own original work and are accurate and honest to the best of my knowledge.

6) I certify that I will disclose any academic integrity findings of responsibility or pending investigations or violations of the Student Code to the Director of the Office of Undergraduate Research and Fellowships prior to applying or seeking nomination for any award. Further, I hereby give permission to the Harvard University Office of Academic Integrity and Student Conduct to release information to the Director of the Office of Undergraduate Research and Fellowships, from official University records

pertaining to academic integrity and conduct offense violations, and to provide contextual information as to the severity of these offenses. The purpose of this release is for Harvard University to consider my application for a merit-based scholarship or fellowship. I have a right to receive a copy of any written information upon request; and, this consent shall remain in effect until the end of the current academic year or until revoked by me in writing, delivered to the Director of the Office of Undergraduate Research and Fellowships, but that any such revocation shall not affect disclosures made prior to the individual's receipt of my written revocation.

7) I affirm my adherence to The Harvard College Honor Code (below):
Members of the Harvard College community commit themselves to producing academic work of integrity – that is, work that adheres to the scholarly and intellectual standards of accurate attribution of sources, appropriate collection and use of data, and transparent acknowledgement of the contribution of others to their ideas, discoveries, interpretations, and conclusions. Cheating on exams or problem sets, plagiarizing or misrepresenting the ideas or language of someone else as one's own, falsifying data, or any other instance of academic dishonesty violates the standards of our community, as well as the standards of the wider world of learning and affairs.

*Signature	Cyrus Urheim 1/31/2026 5:25 PM

HSURV Program Award Acceptance Agreement

Harvard College Office of Undergraduate Research and Fellowships

HSURV Residential Research Program Award Acceptance Agreement

I hereby accept a Harvard College Office of Undergraduate Research and Fellowships ("URAF") Award as a participant in the Program for Research in Science and Engineering (the "Program") for Summer 2026. I understand that I will engage in full-time summer research under the supervision of a Harvard faculty mentor or approved research supervisor, and that the scope of my research activities (the "Project") will be determined in consultation with my mentor and the Program.
By signing below, I acknowledge and agree as follows:

1. Program Period and Full-Time Commitment

1.1 The Program runs from June 3 through August 9, including required move-in, orientation, and community programming as designated by URAF. Staggered move-in is scheduled for June 3–5, with required community programming that weekend. Research begins Monday, June 8 and concludes on Friday, August 7. Fellows should plan to move out between August 8-9.

1.2 The Program is designated as a full-time summer experience. During the funded period, I shall not concurrently participate in other research, internships, employment, coursework, public service projects, or test preparation without prior written approval from URAF.

1.3 Full and active participation in both research and community programming is an essential condition of this Award.

2. Residency Requirement

2.1 I shall reside in Harvard-designated housing within the Harvard Summer Undergraduate Research Village ("HSURV") for the duration of the Program unless expressly approved otherwise by URAF.

2.2 I shall comply with all residential community standards, housing policies, and Harvard University conduct expectations.

2.3 Failure to comply with residential or community standards may result in removal from housing and/or dismissal from the Program.

3. Use of Funds

3.1 Award funds, including stipend, housing, meal benefits, or travel support (if applicable), shall be used solely for purposes approved by URAF and only during the Program period.

3.2 I shall disclose all other sources of summer funding related to the Project.

3.3 I may not receive both wages and a stipend for conducting the same Project.

3.4 Any request to modify the use of Award funds must be submitted to URAF in writing and is subject to approval in URAF's sole discretion.

4. Payment and Installments

4.1 If the Award is paid in installments, continued disbursement is contingent upon satisfactory participation in research and community activities and completion of required compliance obligations.

4.2 URAF reserves the right to delay, reduce, or withhold payment if participation or compliance requirements are not met.

5. Mentor Placement and Research Supervision

5.1 Participation in the Program is contingent upon confirmed placement with a qualified research mentor.

5.2 URAF reserves the right to modify research placements or reassign mentors if necessary to support the objectives of the Program.

5.3 I shall conduct research in accordance with the guidance of my mentor and applicable institutional policies.

6. Reporting and Compliance Obligations

6.1 I shall complete all required onboarding, tax documentation, and payment setup materials by deadlines established by URAF.

6.2 I shall complete all compliance requirements applicable to the Project and Program, including, as applicable:

(a) laboratory and institutional safety training;

(b) IRB approval, exemption, or confirmation of non-human-subjects research;

(c) travel safety requirements (if applicable);

(d) health insurance and immunization requirements applicable to residential students.

6.3 I shall complete all required reflections, progress updates, reports, and deliverables in the form and by the deadlines established by URAF.

6.4 I shall promptly notify URAF of any material changes affecting my ability to participate fully in the Program, including but not limited to changes in mentor placement, research scope, funding, residency status, or other circumstances that may impact my participation.

Failure to satisfy compliance or reporting requirements may result in delayed, reduced, or forfeited funding.

7. Program Participation and Presentations

7.1 I shall participate in required Program components, including orientation, workshops, seminars, community events, and final research presentations.

7.2 I acknowledge that presentations may be recorded and shared publicly.

8. Acknowledgment of Support

8.1 I shall acknowledge Harvard University funding in all publications, presentations, posters, and media appearances relating to the Project.

8.2 The appropriate acknowledgment language will be provided by URAF.

9. Permissions and Records Retention

9.1 I grant Harvard University permission to use my biographical information and participation in the Program for publicity purposes.

9.2 I consent to the retention by URAF of my application materials, reports, presentations, recordings, and research products in accordance with University policy.

10. Reservation of Rights

URAF reserves the right, in its sole discretion, to discontinue, modify, withhold, or require partial or full repayment of Award funds if such action is necessary:

(a) due to my failure to comply with the terms and conditions of the Award or Program;

(b) to protect the purpose and objectives of the Program or URAF; or

(c) to comply with applicable law or Harvard University policy.

Acceptance of this Award does not constitute an employment contract or guarantee of continued appointment beyond the Program period.

*Please sign here:	Cyrus Urheim 3/15/2026 5:45 PM

Request Details

Section Name	Requestee Name	Requestee Email
Academic Evaluator Form Request	Christopher Stubbs	stubbs@g.harvard.edu

PRISE Research Interests Essay

Sonar for Astrophysics: The BATPING (Bidirectional Acoustic Traveltime Pulser for Index Gradients) Project

Cyrus A. Urheim, supervised by Professor Christopher W. Stubbs

The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) will generate an unprecedented catalog of astronomical data to enable precise measurements of the mysterious dark matter and dark energy that together account for 95% of the universe. Understanding their nature is the forefront mission within astrophysics and cosmology today. LSST achieved First Light in July 2025 and now generates 20 terabytes of digital imagery nightly. Our ability to use these data to answer the fundamental physics questions posed by dark matter and dark energy rests on the quality of the LSST images.

Professor Christopher Stubbs has been a longtime leader of LSST/Rubin during its design, construction and commissioning phases. One recent focus of the Stubbs group has been to address image degradation in LSST data due to variable physical conditions within the dome enclosure and near-atmosphere.

I have been working with Professor Stubbs and his group on projects in image quality analysis over the past year, namely: (1) computer vision-based detection and prediction of aircraft contrail streaks from summit sensor data; and (2) supplying the Auxiliary Telescope's photometric calibration system with a Raspberry Pi-based coordinate-finding module. Continuing on this theme, Professor Stubbs and I have identified a new project that we call BATPING (Bidirectional Acoustic Traveltime Pulser for INdex Gradients).

BATPING will allow us to characterize temperature fluctuations within the telescope enclosure that perturb the optical path, decreasing image quality. My work will follow two thrusts: (I) implementation of BATPING's array of acoustic thermometers, and (II) development of a machine learning (ML) approach to optimization of the acoustic signal.

The project's hardware apparatus consists of a piezoelectric transmitter, which propagates an ultrasonic pulse that is measured by two acoustic thermometers positioned on either end of the path. In configuring BATPING's array of transducers, I will benefit from my exposure to acoustic-optic experimentation in Physics 15c "Wave Phenomena": for my lab project last semester, I built an array of microphones to perform acoustic beamforming. Both projects involve the Generalized Cross-Correlation, an algorithm which utilizes Fast Fourier Transforms (FFTs) to determine time delay between correlated waveforms.

The exposure to constructing efficient algorithms I am gaining in CS 1240 will aid my development of an ML-based algorithm to optimize the transmitted waveform. The impulse response of the receiving transducer introduces to the transmitted waveform a "ringdown" tail. The model would learn the transducer impulse response and optimize the sharpness of the output signal peak, enabling adaptive cancellation of the ringdown tail. The Python library PyTorch and add-on torchaudio provide tools to pursue this process.

In addition to Prof. Stubbs, I will work with Prof. Paul Horowitz on electronics for BATPING, and with post-doctoral researcher Dr. Johnny Esteves on harnessing ML methods.

Through image quality assurance efforts like BATPING, it is exciting to think I will be enhancing the science reach of Rubin/LSST in some way. In the process, I will also gain a

deeper understanding of image fidelity that will subsequently allow me to contribute to my own scientific analyses of LSST data.

Works Cited

Kurmus, A., et al. “Directly Characterizing Dome Seeing: Differential Image Motion Sensor Using Multisources (DIMSUM).” arXiv, arXiv:2311.03692, 2023.

Marangola, M. A., et al. “Measuring Local Turbulence Along the Optical Path: Multi-Beam Optical Seeing Sensor (MOSS).” arXiv, arXiv:2508.00678, 2025.

Sjoberg, K., et al. “PANDORA: New Avenues in Sub-Percent Photometric Precision for Ground-Based Astronomy.” American Astronomical Society Meeting Abstracts, vol. 245, p. 302.21, 2025.

Sjoberg, K., et al. “PANDORA: New Avenues in Sub-Percent Photometric Precision for Ground-Based Astronomy.” American Astronomical Society Meeting Abstracts, vol. 245, p. 302.21, 2025.

Vera C. Rubin Observatory Team. RTN-095: The Vera C. Rubin Observatory Data Preview 1. NSF–DOE Vera C. Rubin Observatory, 2025.

Vogiatzis, K., et al. “Dome Seeing Sensitivity Analysis for LSST.” Modeling, Systems Engineering, and Project Management for Astronomy VIII, vol. 10705, 1070503, 2018.

Zago, L. “Engineering Handbook for Local and Dome Seeing.” Optical Telescopes of Today and Tomorrow, vol. 2871, pp. 726–736, 1997.

PRISE Community Essay

Cyrus A. Urheim

In my first three semesters, I have engaged in multiple communities where leadership is delocalized and distributed across the cohort of students. This spirit is most prominent in the Harvard Bach Society, a fully student-run organization that conducts, manages, and finances itself. In this setting, every member is called into action to select repertoire for future performances, coordinate rehearsals, and, more generally, bring an unbridled love for the music that exists without a traditional conductor imposing deadlines and standards. As principal violist in the orchestra, I find myself more aware of my communication with others, through both words and music. In this setting, I feel more emboldened to check in on my standmate when I sense that he's distracted, or to approach one of the flutists that I don't have any classes with. These experiences help me envision the community that I would like to create during PRISE.

I aim to bring to HSURV my respect and love for disciplines ranging from sociology to physics, but also my experiences in the strengthening of communities themselves. I am excited to join the cohort of PRISE researchers whose interests, while distinct, are interconnected. My research experiences prior to Harvard in two widely different areas – computational sociology and physical chemistry – has broadened the scope of my current interest in astrophysics and motivates me to engage with others working in areas unfamiliar to me. Related to this, I would be interested in forming an interdisciplinary journal club with my PRISE cohort, sharing and catalyzing ideas between our disciplines.

This community-building across disciplines is a deliberate, thoughtful process. I have grown to appreciate the importance of leading conversation and driving connections as a

co-director of the Harvard Best Buddies Program, where we build friendships with people with intellectual and developmental disabilities in the Cambridge community. I see residents of a specialized housing program every week, and especially enjoy coffee runs with one of the residents, Amy. Making our time together meaningful means engaging my friend, who has speaking limitations, in creative and expansive ways, while respecting the boundaries of our conversations. With this in mind, group trips to cafes near campus, or walks and runs along the Charles river, would be meaningful and enjoyable ways to connect without placing unmanageable expectations on others.

In whatever capacity community at PRISE develops, I hope that open-mindedness and commitment to collective experience remains a primary goal. As a proud representative of Mather IMs, I have found myself in sports both familiar (soccer) and unfamiliar (volleyball). Some of the other soccer players from Harvard's club team treated IM games like club games, opting to exclude people who lacked similar experience. Remembering how nerve-racking it was to step in for Mather's A team at the volleyball tournament, I felt contentment in our unity, especially in losses.

Cyrus Aziz Urheim

Cambridge, MA | cyrusurheim@college.harvard.edu | +1 (812) 837-3358 | GitHub

Education

Harvard University, *Joint A. B. in Chemistry and Physics, Computer Science, Secondary in Sociology* Sept 2024 – May 2028

- **Relevant Coursework:** PHYS16: Mechanics and Special Relativity, CS50: Computer Science, LS1b: Genetics, Genomics, Evolution, CHEM20/30: Advanced Organic Chemistry I and II, PHYS15b: Electromagnetism, PHYS15c: Waves Phenomena, STAT110: Advanced Probability
- **Activities:** Harvard Bach Society Orchestra (principal violist), Harvard-Radcliffe Orchestra (violinist), PBHA Best Buddies Volunteer Program (Director), Harvard-Radcliffe Society of Physics Students (Events and Panels), Harvard Climbing Team, Harvard Club Futsal

Indiana University, *Dual Enrollment Student* Aug 2022 – May 2024

- **Coursework:** MATH-M311: Multivariate Calculus (A+), M312: Vector Calculus (A+), S413: Honors Real Analysis I (A), S414: Honors Real Analysis II (A+), CHEM-C341: Organic Chemistry I (A)

Bloomington High School North, *Honors Diploma* Aug 2020 – May 2024

- **GPA:** 4.65, **SAT:** 1570, 12 Advanced Placement Courses (all 5's), National Merit Scholar
- **Activities:** Varsity Soccer (Captain, All-District), Quiz Bowl (Founder), Orchestra (Principal Violist)

Research Experience

Computational Astrophysics Research Intern, *Christopher Stubbs Lab, Harvard Department of Physics* Feb 2025 – Present

- Conducting research in the Vera Rubin Observatory's Legacy Survey of Space and Time (LSST) Project, focused on precision cosmology and image quality analysis
- Received funding from the Harvard College Research Program to build a supervised learning algorithm for detection and prediction of aircraft contrails in LSST data

J-Term Independent Researcher, *Supervised by Professor Howard Georgi, Harvard Department of Physics* Dec 2024 – Feb 2025

- Research on the extreme Estonian sport of Kiiking as a continuation of the final project in Prof. Georgi's class, Physics 16
- Mathematically modeled the physical system of Kiiking, producing an optimized strategy of kinetic motion (report here)

Physical Chemistry Research Intern, *Dragnea Lab, Indiana University Dept. of Chemistry* May 2023 – May 2024

- Worked on project studying superradiance in a system of multifluorophore-conjugated Brome Mosaic Virus (BMV) particles under pulse excitation by laser light
- Assisted in TIRF (Total Internal Reflection Fluorescence) microscopy of Brome Mosaic Virus samples that show brightening due to collective dynamics of capsid shell
- Analyzed radiation brightening of virus scaffolds using ImageJ software to track pixel values in microscope image sequences

Computational Sociology Research Intern, *Rojas Group, IU Department of Sociology* May 2023 – Dec 2024

- Built Python code to analyze bibliometric data and text from archived newspapers and journals to analyze the life and legacy of W.E.B. Du Bois (abstract here)
- Natural Language Processing Tools: Hugging Face Transformers (GPT-based sentiment analysis), SpaCy (Named Entity Disambiguation), BERTopic (Topic Clustering), Selenium (webscraping)

Awards

National Merit Scholar — Jan 2024

"Mathematics of Fractals" High School Competition: Third Place, *IUPUI Math Department* — May 2023

AP Chemistry Outstanding Student Award, *Bloomington High School North* — May 2023

Indiana Science Olympiad, *Detector Building (3rd Place), Cellular Biology (3rd Place)* — May 2023

AP Physics C Outstanding Student Award, *Bloomington High School North* — May 2022

Presentations

Indiana University Political Science Workshop, *The Case of W.E.B. Du Bois and a Sequential Theory of Canonization* (here) — Nov 2023

- Presenters: Eehyun Kim, Tania Ravaei, Cyrus Urheim

American Sociological Association Annual Meeting, *From "Scholar Denied" to "Architect of the Civil Rights Movement": Lessons from W.E.B. Du Bois's Canonization* — Aug 2024

- Presenters: Eehyun Kim, Tania Ravaei (presenter), Cyrus Urheim, Tiffany Vo

Work Experience

Cashier, *Kroger Supermarket-1175 S College Mall Rd, Bloomington, IN* May 2024 – Sept 2024

- Part-time; bagged groceries, returned carts from parking lot to in-store kiosks

Volunteer, *Concord Ave Assisted Living, Cambridge, MA* Sept 2024 – Present

- Previously worked at Stone Belt Center for Developmental Disabilities (Bloomington, IN)
- Current Director of the Harvard Best Buddies Chapter
- Assist with support programs for individuals with intellectual and developmental disabilities

Writing Tutor, *Harvard College Writing Center* Sept 2025 - Present

- Tutor Harvard College students on structure and clarity of writing in hour-long, one-on-one meetings for all disciplines within the College ((profile)

Skills

Software and Computing: Python (NumPy, Pandas, MatplotLib, Scikit, PyTorch, TensorFlow, NLTK), C, Java, SQL, JavaScript, HTML, LaTeX, Mathematica

Tools: ImageJ (Fiji), IGOR Pro, TIRF Microscopy, Excel, Confluence

Music: Violin (IU Jacobs School of Music Pre-College Program alumnus), Viola, Kamancheh

Languages: English, German (3 years), French (2 years), Farsi (1 year)

Harvard University

Cambridge, Massachusetts 02138
Harvard College

Urheim, Cyrus Aziz
Admitted in 2024
Good Academic Standing

Mather House
HUID: 31712289

Academic Program

Concentration: Chemistry and Physics

Beginning of Harvard College Record

2024 Fall

Course	Description	Earned	Grade
COMPSCI 50	Introduction to Computer Science	4.000	SAT
FYSEMR 32R	Autobiography and Black Freedom Struggles	4.000	SAT
GENED 1034	Texts in Transition	4.000	A-
PHYSICS 16	Mechanics and Special Relativity	4.000	A

2025 Spring

Course	Description	Earned	Grade
CHEM 20	Organic Chemistry	4.000	A
EXPOS 20	Expository Writing 20	4.000	A
Course Topic:	The "Science" of Science Ficti		
LIFESCI 1B	An Integrated Introduction to the Life Sciences: Genetics, Genomics, and Evolution	4.000	A
PHYSICS 15B	Introductory Electromagnetism	4.000	A

2025 Fall

Course	Description	Earned	Grade
CHEM 30	Organic Chemistry	4.000	A
PHYSICS 15C	Wave Phenomena	4.000	A
SOCIOL 1101	Inequality Then and Now: The History of Human Suffering	4.000	A
STAT 110	Introduction to Probability	4.000	A

2026 Spring

Course	Description	Earned	Grade
CHEM 27	Organic Chemistry of Life		
COMPSCI 1240	Data Structures and Algorithms		
GENED 1051	Reclaiming Argument: Logic as a Force for Good		
PHYSICS 143A	Quantum Mechanics I		
STAT 111	Introduction to Statistical Inference		

Harvard College Career Totals

Cum GPA:	3.967	Cum Totals	48.000	40.000

End of Harvard College Record

UNOFFICIAL

Date Issued: Feb 1, 2026

Erika J. McDonald, Registrar
Not official unless signed and sealed